

DRAEM - ICCV 2021

A Discriminatively Trained Reconstruction Embedding for Surface Anomaly Detection – Vitjan Zavrtanik, Matej Kristan, Danijel Skočaj

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- Detection and localization of anomalies in the image space
- E.g. RGB drone images, medical scans, ...
- OOD samples are often rare and diverse
- Collecting data for all possible anomalies is often infeasible
- Data collection gets even harder when pixel-accurate localization is required
- Therefore, methods that only rely on normal data are preferred

Problem Statement

Current Situation and Related Work

- Two groups of methods in this area:
 1. **Reconstructive Methods** (Autoencoders, GANs) detect anomalies by measuring the reconstruction error
 - Minor anomalies are often well reconstructed
 2. **Discriminative Methods** detect distribution deviations on (deep) features (e.g. from CNNs)
 - Often suffer from blurry or coarse segmentation
 - Overfit to synthetic anomalies

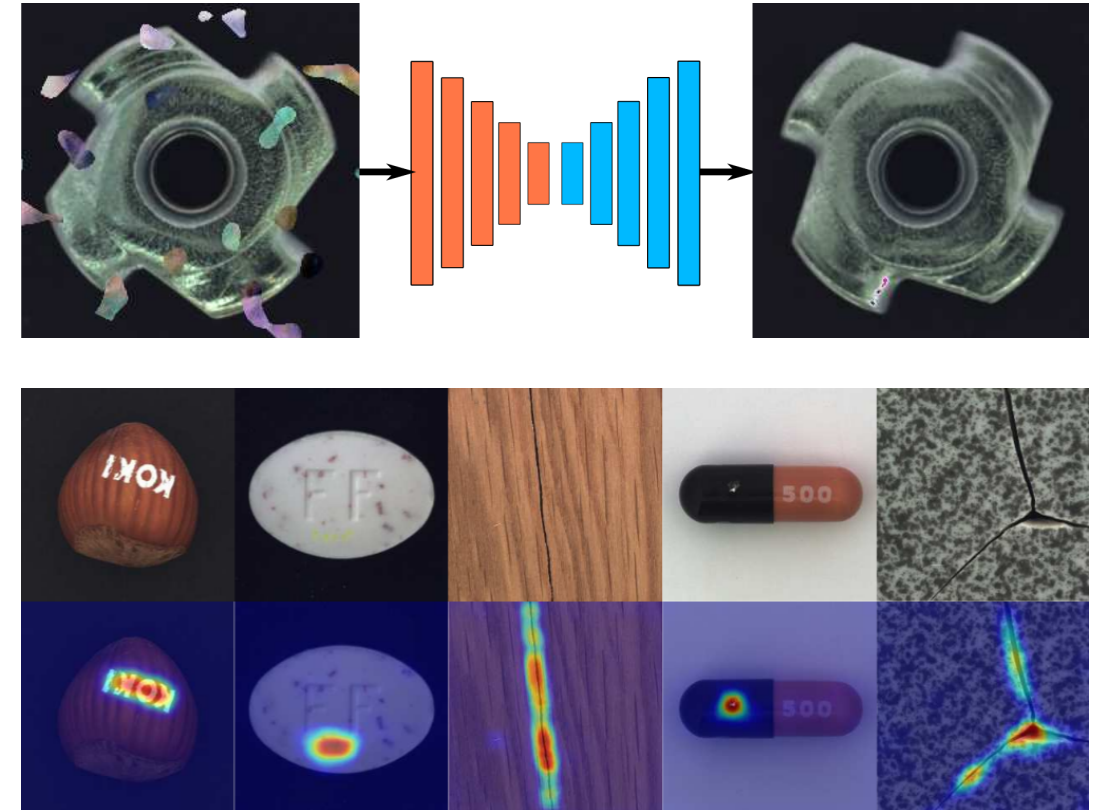
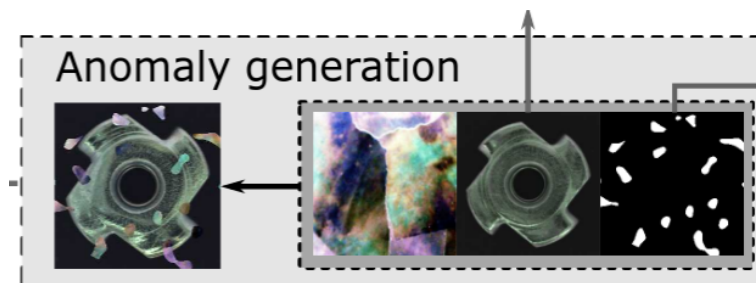


Figure: Blurry segmentation maps from discriminative methods.

Framework Overview

How it Works



Performance Comparison

DRÆM vs State-of-the-Art

Method	Image AUROC ↑	Pixel AUROC ↑	Pixel AP ↑
US	87.7%	93.9%	45.5%
RIAD	91.7%	94.2%	48.2%
PaDim	<u>95.5%</u>	97.4%	<u>55.5%</u>
DRÆM	98.0%	<u>97.3%</u>	68.4%

Table: Results on the MVTec dataset.

Performance on DAGM Dataset

Unsupervised vs. Supervised Methods

Method	AUROC ↑	TPR ↑	TNR ↑	CA ↑
Unsupervised				
US	72.5	72.6	65.3	66.2
RIAD	78.6	79.2	69.1	70.4
PaDim	95.0	83.3	97.5	95.7
DRÆM	99.0	96.5	99.4	98.5
Supervised				
CADN	—	—	—	89.1
Racki et al.	99.6	99.9	99.5	—
Lin et al.	99.0	99.4	99.9	—
Bozic et al.	100	100	100	100

Table: DRÆM on DAGM dataset: outperforming unsupervised methods and on par with fully supervised ones.

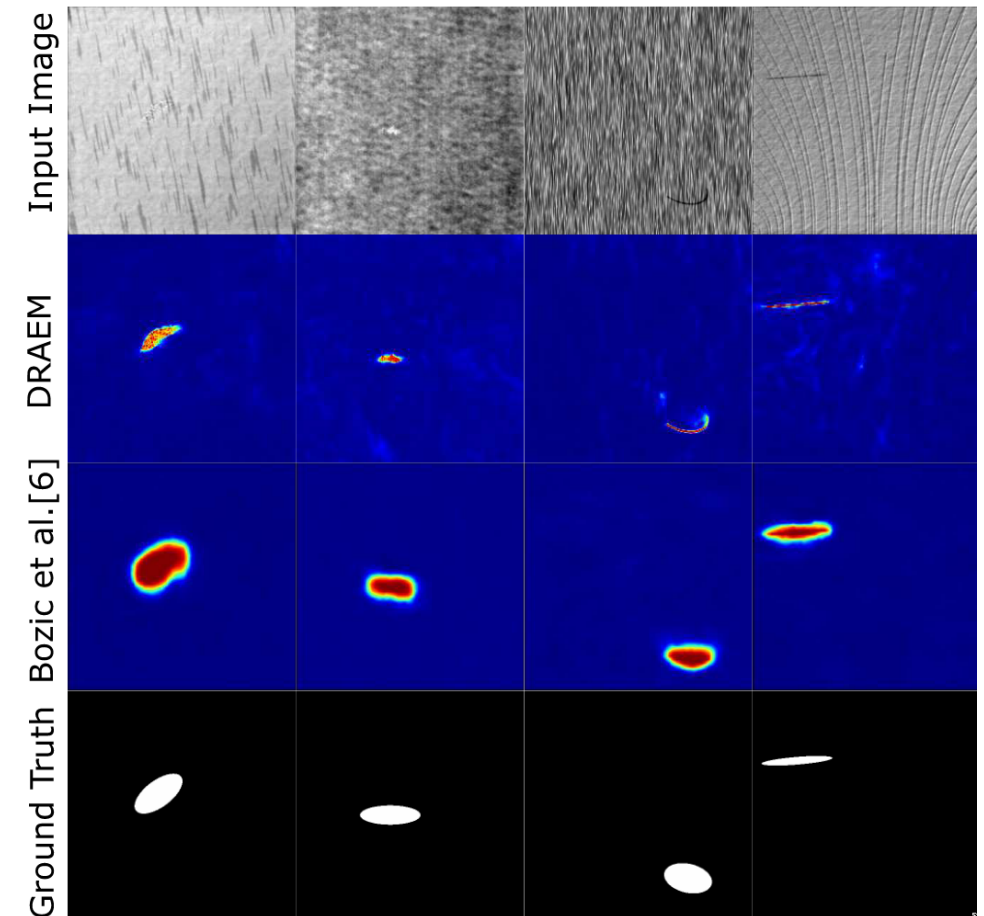
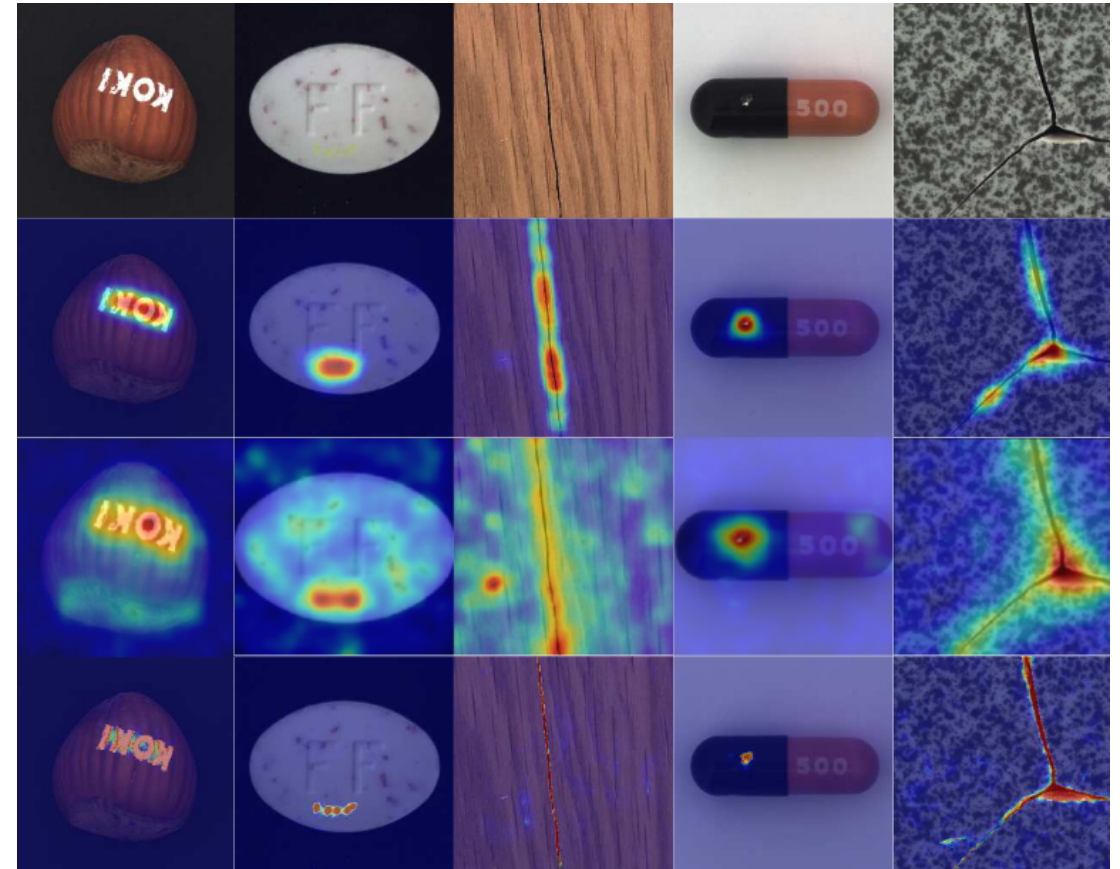


Figure: DRÆM vs supervised SOTA (Bozic et al.) on the DAGM dataset.

Key Findings & Application & Conclusion

The Value of DRÆM

- Outperforms unsupervised SOTA methods
- Similar performance to supervised SOTA methods
- High-quality segmentation can help downstream tasks
 - Estimate the size of the impacted area
 - Distinguish between cracks, corrosion, or just paint
- We don't need expensive hand-labeled data
- We only need just-out-of-distribution patterns for anomaly generation



The Provided Code

Complete and Good?

- × No code for experiments on the DAGM dataset
- × Inconsistencies in variable meaning between code and paper (e.g., β as anomaly opacity vs. image opacity)
- × Incorrect weighting of incomplete batches
- ✓ Complete code for training and evaluation (for main findings)
- ✓ Well-structured code in separate files for models, loss, data loading, training and evaluation
- ✓ Well readable code
- ✓ Download script for MVTec and DTD datasets

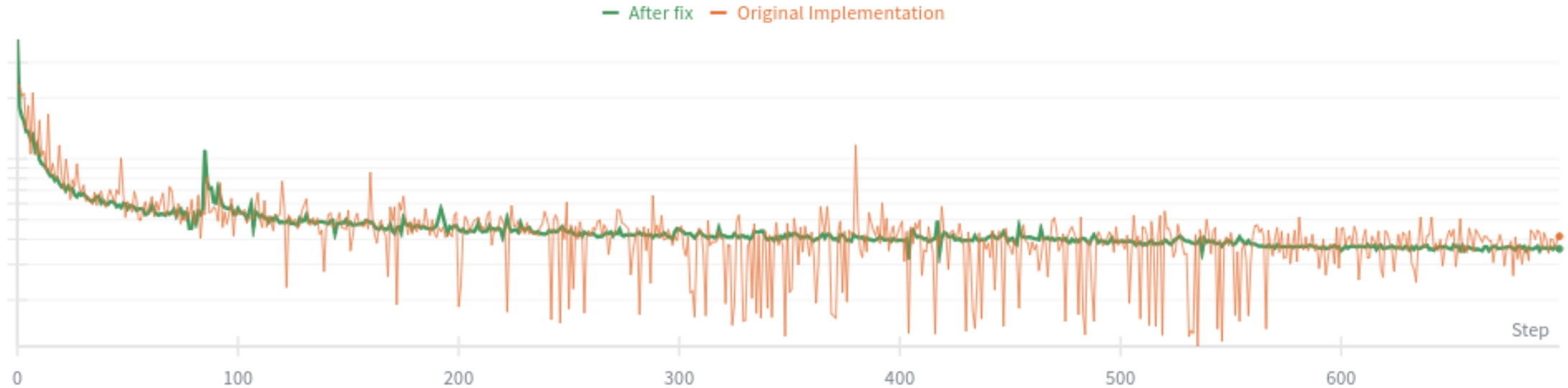


Figure: Loss smoothing by weighting the last batch of each epoch.

- Ablation studies in the paper show that adding transparency to anomalies has a huge impact
- The original implementation samples a single $\beta \in [0.1, 1]$ for the whole image (uniform β)
- I added four different blending methods:

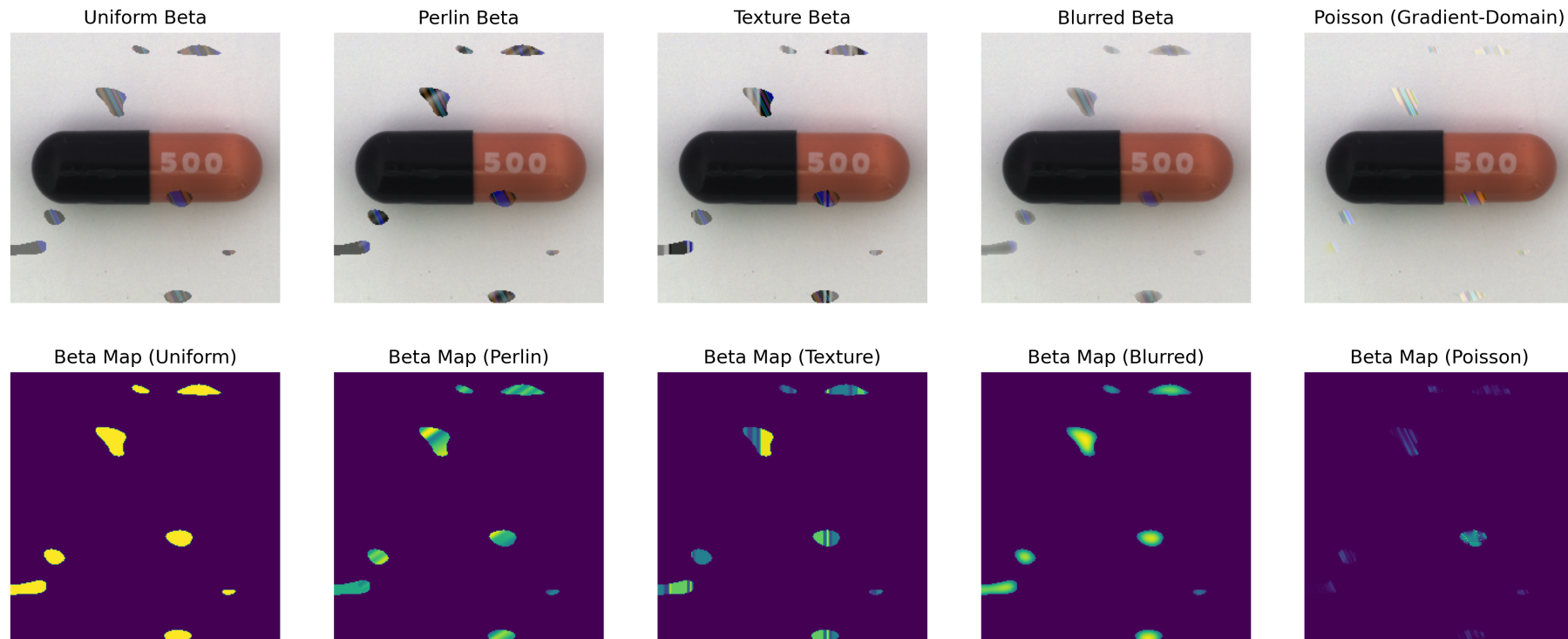
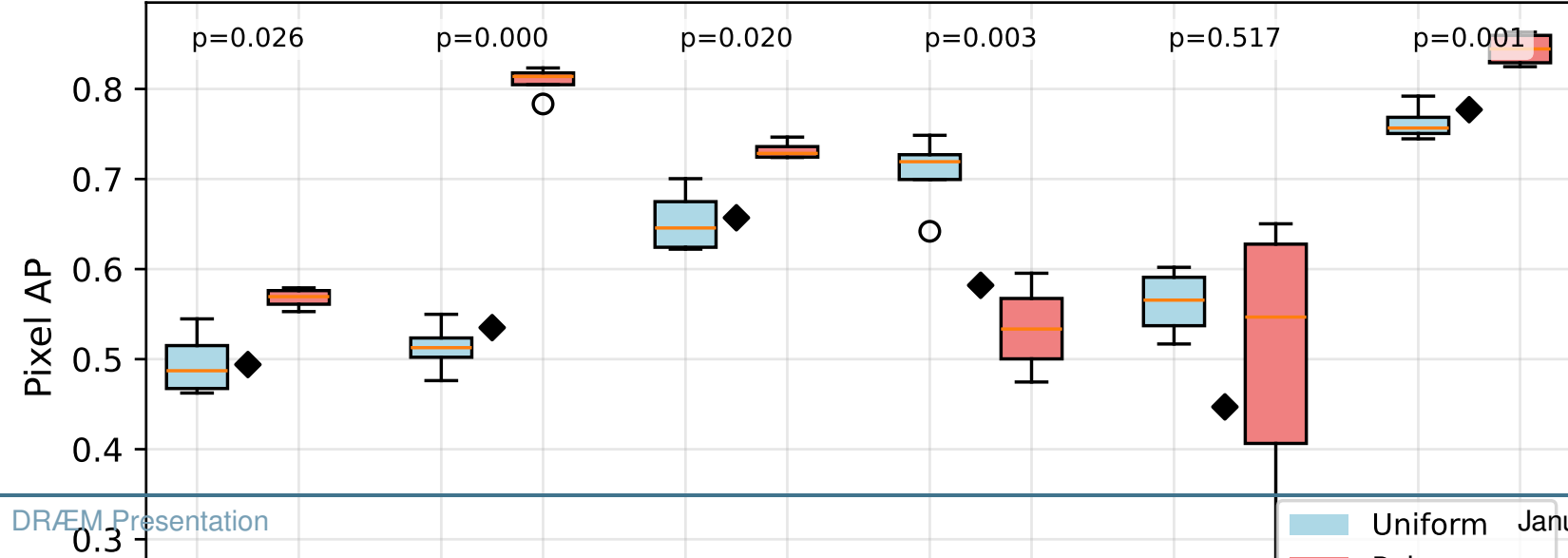


Figure: Different blending methods (from left to right): uniform β , Perlin- β , texture- β , blurred β , Poisson Image Interpolation.

Method	Image AUROC $\uparrow$	Pixel AUROC $\uparrow$	Pixel AP $\uparrow$
DRÆM-Uniform	98.0%	97.3%	68.4%
DRÆM-Blurred	96.4%	96.0%	65.8%
DRÆM-Texture	98.0%	97.2%	69.1%
DRÆM-Perlin	98.2%	97.3%	69.5%
DRÆM-Poisson	<u>99.1%</u>	<u>97.8%</u>	<u>71.6%</u>
DRÆM-Poisson+Perlin	99.2%	98.0%	73.6%

Table: Results on the MVTec dataset for different blending methods.



Thank you for your attention.

The background of the slide is a solid blue color. In the lower half, there is a decorative graphic consisting of numerous thin, concentric, curved lines that create a sense of motion or a ripple effect, starting from the bottom and curving upwards towards the center.